Validation engine - proof report

PClaimAssist research tests, Objective 3 (iii), Eq. 3.7-3.10 - results of 2026-11-05 18:57:52, report generated 2026-11-05

Result. 40 errors were seeded into 20 records and 20 records were left clean. The validation engine flagged **40 of 40** seeded errors: precision **1.0**, recall **1.0**, F1 **1.0**; false alarm rate **0.0** (0 of 20 clean records flagged). **All 31 rules are covered**: 27 by seeded errors on real records, 4 by an engine-level check (they cannot be reached on a real claim - section 3).

Read with care. The errors were designed from the rule definitions, at least one per rule the engine can reach, so the result shows the rules work as written - not how many of all real-world claim errors they would catch. The pictures are the page's own "Rule Check Results" panel for each record, as staff see it.

Contents

1. Result at a glance
2. How the test works
3. Every rule and how it is covered
4. The seeded errors
5. Every seeded record and what the engine flagged
6. Engine-level checks (IC-01, IC-02, IC-07, MF-08)
7. The clean records
8. What this means

The PDF also has bookmarks, and every page links back to these contents.

1. Result at a glance

Measure	Value
Records (clean / seeded)	20 / 20
Seeded errors	40
Seeded errors detected	40
True positives (TP)	40
False positives (FP)	0
False negatives (FN)	0
Precision (Eq. 3.7)	1.0
Recall (Eq. 3.8)	1.0
F1 score (Eq. 3.9)	1.0
Clean records with a flag	0 of 20
False alarm rate (Eq. 3.10)	0.0

Per rule category

Rule category	Seeded errors (expected flags)	TP	FP	FN	Precision	Recall	F1
Missing Field	15	15	0	0	1.0	1.0	1.0
Invalid Format	7	7	0	0	1.0	1.0	1.0
Inconsistent Entry	9	9	0	0	1.0	1.0	1.0
Signature	6	6	0	0	1.0	1.0	1.0
Image Quality	1	1	0	0	1.0	1.0	1.0
Correction Status	2	2	0	0	1.0	1.0	1.0
All rules	40	40	0	0	1.0	1.0	1.0

Per seeded category

Seeded category	Errors	Detected	Missed	Detection rate
Missing CF4 field	4	4	0	1.0
Inconsistent entry	9	9	0	1.0
Signature	6	6	0	1.0
Missing field	11	11	0	1.0
Image quality / correction status	3	3	0	1.0
Invalid format	7	7	0	1.0
All seeded errors	40	40	0	1.0

Unit: one rule on one record. R = rules the record's seeded errors must fire, A = rules they may also fire as a direct consequence (e.g. a blanked field adds to the "more than 3 blank" count), F = rules fired. TP = |F and R|, FN = |R - F|, FP = |F - R - A|. Precision = TP/(TP+FP) (Eq. 3.7), recall = TP/(TP+FN) (Eq. 3.8), F1 = 2PR/(P+R) (Eq. 3.9), false alarm rate = clean records with any flag / clean records (Eq. 3.10).

2. How the test works

40 records are built from the answer key's fictional patients: each patient's confirmed values go through the real auto-population into the real forms page (offline, in Microsoft Edge), plus the fictional details staff would type on the forms. 20 records are left clean; 20 get two seeded errors each (40 in all: every error kind once, the commonest kinds twice). Each error is typed where staff would type it, and placed only on a record where its rule applies (a CF4 rule on a claim with CF4, a delivery rule on a record with a delivery); two errors in one record never touch the same field. The page's own validation (validation-rules.js runValidationRules, as the page runs it) is then run, with the image-quality, correction and signature checks switched on, and its flags compared with the errors seeded.

The test also checks that the page kept each seeded value: a value the page itself puts back cannot be detected and would not be a fair error. Every seeded value in this run was kept.

3. Every rule and how it is covered

Rule	Category	Action	Seeded	Caught	Covered by	Result
IQ-01	Image Quality	display_warning	1	1	seeded end to end	1 of 1 caught
CR-01	Correction Status	display_warning	1	1	seeded end to end	1 of 1 caught
CR-02	Correction Status	flag_high_risk	1	1	seeded end to end	1 of 1 caught
MF-01	Missing Field	block_submission	2	2	seeded end to end	2 of 2 caught
MF-02	Missing Field	block_submission	2	2	seeded end to end	2 of 2 caught
MF-03	Missing Field	display_warning	2	2	seeded end to end	2 of 2 caught
MF-04	Missing Field	block_submission	2	2	seeded end to end	2 of 2 caught
MF-05	Missing Field	block_submission	2	2	seeded end to end	2 of 2 caught
MF-06	Missing Field	flag_high_risk	1	1	seeded end to end	1 of 1 caught
MF-07	Missing Field	block_submission	2	2	seeded end to end	2 of 2 caught
MF-08	Missing Field	block_submission	0	0	engine-level check	fires, and only it
MF-09	Missing Field	display_warning	2	2	seeded end to end	2 of 2 caught
IF-01	Invalid Format	block_submission	2	2	seeded end to end	2 of 2 caught
IF-02	Invalid Format	block_submission	2	2	seeded end to end	2 of 2 caught
IF-03	Invalid Format	block_submission	2	2	seeded end to end	2 of 2 caught
IF-04	Invalid Format	display_warning	1	1	seeded end to end	1 of 1 caught
IC-01	Inconsistent Entry	block_submission	0	0	engine-level check	fires, and only it
IC-02	Inconsistent Entry	block_submission	0	0	engine-level check	fires, and only it
IC-03	Inconsistent Entry	block_submission	2	2	seeded end to end	2 of 2 caught
IC-04	Inconsistent Entry	block_submission	2	2	seeded end to end	2 of 2 caught
IC-05	Inconsistent Entry	display_warning	1	1	seeded end to end	1 of 1 caught
IC-06	Inconsistent Entry	display_warning	1	1	seeded end to end	1 of 1 caught
IC-07	Inconsistent Entry	block_submission	0	0	engine-level check	fires, and only it
IC-09	Inconsistent Entry	block_submission	1	1	seeded end to end	1 of 1 caught
IC-08	Inconsistent Entry	block_submission	2	2	seeded end to end	2 of 2 caught
SP-01	Signature	block_submission	1	1	seeded end to end	1 of 1 caught
SP-02	Signature	block_submission	1	1	seeded end to end	1 of 1 caught
SP-03	Signature	block_submission	1	1	seeded end to end	1 of 1 caught
SP-04	Signature	display_warning	1	1	seeded end to end	1 of 1 caught
SP-05	Signature	display_warning	1	1	seeded end to end	1 of 1 caught
SP-06	Signature	block_submission	1	1	seeded end to end	1 of 1 caught

Why four rules are checked at engine level

Rule	Why it cannot be seeded on a real claim
MF-08	CF4 course in the ward is auto-filled with a discharge line: the page writes it whenever the course is empty (app.js DERIVED_FIELDS cf4CourseOrder2), so the engine never sees it blank.
IC-01	Consistent by construction: all five forms fill from one entry (the engine fixes name_match_across_forms = 1).
IC-02	Consistent by construction: one PIN feeds every form (pin_match_across_forms = 1).
IC-07	Consistent by construction: one pair of dates feeds every form (dates_match_across_forms = 1).

4. The seeded errors

Every error, the record it was seeded into, exactly what was changed (before -> seeded), the rule it must fire, and whether it did. Click a record to see its panel.

Record	Patient	Category	Error	What was changed (before -> seeded)	Must fire	Flagged
S01	R01	Missing CF4 field	CF4 chief complaint left blank	cf4ChiefComplaint: HYPOGASTRIC PAIN RADIATING AT THE ____ -> (blank)	MF-07	yes
S01	R01	Inconsistent entry	Member born after admission	patientDOB: 1993-01-09 -> 2026-09-28	IC-04	yes
S02	R02	Signature	Attending midwife signature missing on CF4	midwife signature: present -> missing	SP-02	yes
S02	R02	Missing field	Member date of birth left blank	patientDOB: 1997-02-08 -> (blank)	MF-03	yes
S03	R03	Inconsistent entry	Unmarried patient filed under another member (her father), relationship to member left blank	memberIsPatient: ticked -> not ticked; memberFirstName: AILEEN -> ROLANDO; memberLastName: MANGAO -> MANGAO; memberSex: Female -> Male; civilStatus: Married -> Single; relationship: Self -> (blank)	IC-06	yes
S03	R03	Missing field	Date discharged left blank	dateDischarge: 2026-09-29 -> (blank)	MF-05	yes
S04	R04	Image quality / correction status	Logbook page failed the image quality check	image quality check: passed -> failed	IQ-01	yes
S04	R04	Missing field	Member PIN left blank	patientPIN: 99-123488376-4 -> (blank); memberPIN: 99-123488376-4 -> (blank)	MF-02	yes
S05	R05	Inconsistent entry	Delivery date before admission	deliveryDate: 2026-08-29 -> 2026-08-28	IC-05	yes
S05	R05	Invalid format	Birth date kept as unreadable text (OCR's letter O for a zero)	patientDOB: 2026-08-29 -> O8-29-2026	IF-03	yes
S06	R06	Inconsistent entry	Discharged the same day as the delivery	dateDischarge: 2026-09-07 -> 2026-09-06	IC-08	yes
S06	R06	Invalid format	Digit in the patient's first name	patientFirstName: BABY BOY -> B1BY BOY	IF-04	yes
S07	R07	Inconsistent entry	Discharged the same day as the delivery	dateDischarge: 2026-09-29 -> 2026-09-28	IC-08	yes
S07	R07	Invalid format	Letter in the PIN	memberPIN: 99-123512133-7 -> 99-12O3512133-7	IF-02	yes
S08	R08	Missing field	Patient name left blank	patientLastName: CAMINADE -> (blank); patientFirstName: MARY JOY -> (blank); patientMiddleName: SABANAL -> (blank); patientNameExt: (blank) -> (blank)	MF-01	yes
S08	R08	Missing field	Date admitted left blank	dateAdmitted: 2026-08-29 -> (blank)	MF-04	yes
S09	R09	Missing field	Member date of birth left blank	patientDOB: 1990-05-30 -> (blank)	MF-03	yes
S09	R09	Missing field	Four required fields left blank (accommodation, disposition, HCI accreditation no., HCI name)	accommodation: Non-Private -> (blank); disposition: Improved -> (blank); hciPAN: H99999999 -> (blank); hciName: SAMPLE LYING-IN CLINIC -> (blank)	MF-06	yes
S10	R11	Inconsistent entry	Member born after admission	memberDOB: 1999-04-11 -> 2026-09-27	IC-04	yes
S10	R11	Missing field	Member PIN left blank	memberPIN: 99-123543809-1 -> (blank)	MF-02	yes
S11	R12	Missing field	Date admitted left blank	dateAdmitted: 2026-08-29 -> (blank)	MF-04	yes
S11	R12	Invalid format	PIN with 11 digits	memberPIN: 99-123551728-2 -> 99-12355172-8	IF-01	yes
S12	R14	Missing field	Patient name left blank	patientLastName: DAÑO -> (blank); patientFirstName: SHIELA MAE -> (blank); patientMiddleName: ALCUIZAR -> (blank); patientNameExt: (blank) -> (blank)	MF-01	yes

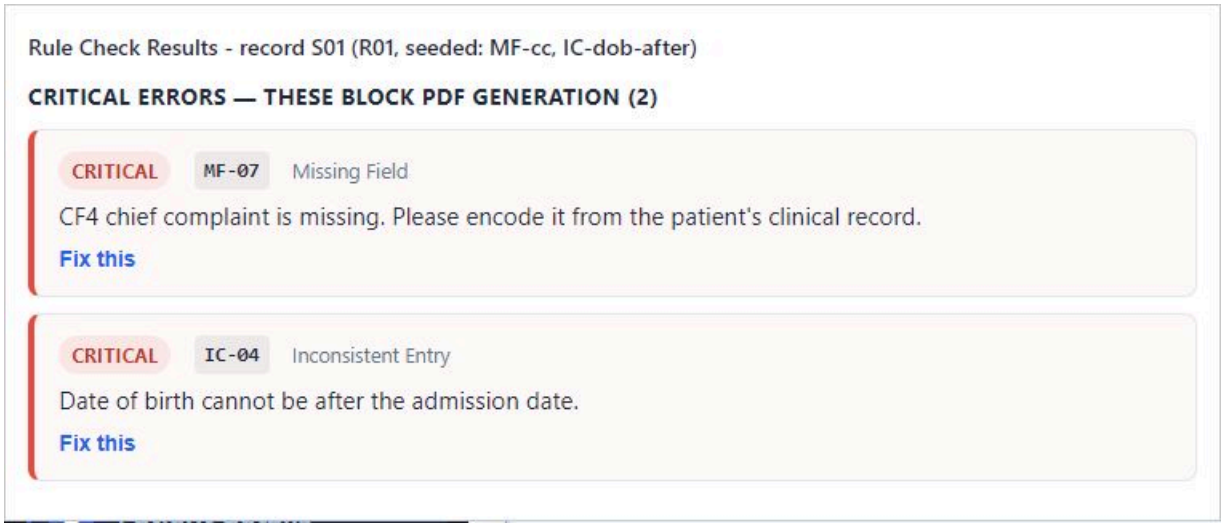
S12	R14	Image quality / correction status	Five correction suggestions not yet reviewed	correction suggestions not reviewed: 0 -> 5	CR-02	yes
S13	R15	Inconsistent entry	Discharge date before admission	dateDischarge: 2026-09-04 -> 2026-09-02	IC-03	yes
S13	R15	Invalid format	Letter in the PIN	patientPIN: 99-123575485-5 -> 99-12O3575485-5; memberPIN: 99-123575485-5 -> 99-12O3575485-5	IF-02	yes
S14	R16	Invalid format	PIN with 11 digits	patientPIN: 99-123583404-6 -> 99-12358340-4; memberPIN: 99-123583404-6 -> 99-12358340-4	IF-01	yes
S14	R16	Signature	Thumbmark without the HCI assisting representative	thumbmark signature: no thumbmark -> thumbmark used; hciAssist signature: present -> missing	SP-05	yes
S15	R17	Missing field	Date discharged left blank	dateDischarge: 2026-08-27 -> (blank)	MF-05	yes
S15	R17	Signature	Member signature missing (signed form checked)	member signature: present -> missing	SP-01	yes
S16	R18	Missing CF4 field	CF4 drugs administered left blank	cf4DrugName1: COMBINED ORAL CONTRACEPTIVE PILLS -> (blank); cf4DrugName2: (blank) -> (blank); cf4DrugName3: (blank) -> (blank)	MF-09	yes
S16	R18	Inconsistent entry	Discharge date before admission	dateDischarge: 2026-08-30 -> 2026-08-28	IC-03	yes
S17	R19	Missing CF4 field	CF4 drugs administered left blank	cf4DrugName1: MEDROXYPROGESTERONE ACETATE (DMPA) -> (blank); cf4DrugName2: (blank) -> (blank); cf4DrugName3: (blank) -> (blank)	MF-09	yes
S17	R19	Signature	HCI representative signature missing	hciRep signature: present -> missing	SP-03	yes
S18	R20	Signature	Attending provider signature missing on CF4	cf4Provider signature: present -> missing	SP-06	yes
S18	R20	Signature	Employer signature missing for an employed member	memberType: (blank) -> Employed Private; employer signature: present -> missing	SP-04	yes
S19	R01	Missing CF4 field	CF4 chief complaint left blank	cf4ChiefComplaint: HYPOGASTRIC PAIN RADIATING AT THE ____ -> (blank)	MF-07	yes
S19	R01	Invalid format	Birth date kept as unreadable text (OCR's letter O for a zero)	patientDOB: 1993-01-09 -> O1-09-1993	IF-03	yes
S20	R02	Inconsistent entry	CF2 and CF4 diagnoses differ	dxADiagnosis: SPONTANEOUS VERTEX DELIVERY -> PREGNANCY UTERINE; cf4FinalDiagnosis: PUFT 39 WKS AOG DELIVERED VIA NSVD TO A LIVE TERM BABY GIRL AGA W/ AS OF 8,9 G2P2 (2002) -> ACUTE GASTROENTERITIS	IC-09	yes
S20	R02	Image quality / correction status	One correction suggestion not yet reviewed	correction suggestions not reviewed: 0 -> 1	CR-01	yes

5. Every seeded record and what the engine flagged

Record S01 - patient R01 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
CF4 chief complaint left blank	cf4ChiefComplaint: HYPOGASTRIC PAIN RADIATING AT THE ____ -> (blank)	MF-07	MF-06
Member born after admission	patientDOB: 1993-01-09 -> 2026-09-28	IC-04	-

Fired: **IC-04, MF-07**. Caught: **IC-04, MF-07**



Record S02 - patient R02 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
Attending midwife signature missing on CF4	midwife signature: present -> missing	SP-02	-
Member date of birth left blank	patientDOB: 1997-02-08 -> (blank)	MF-03	MF-06

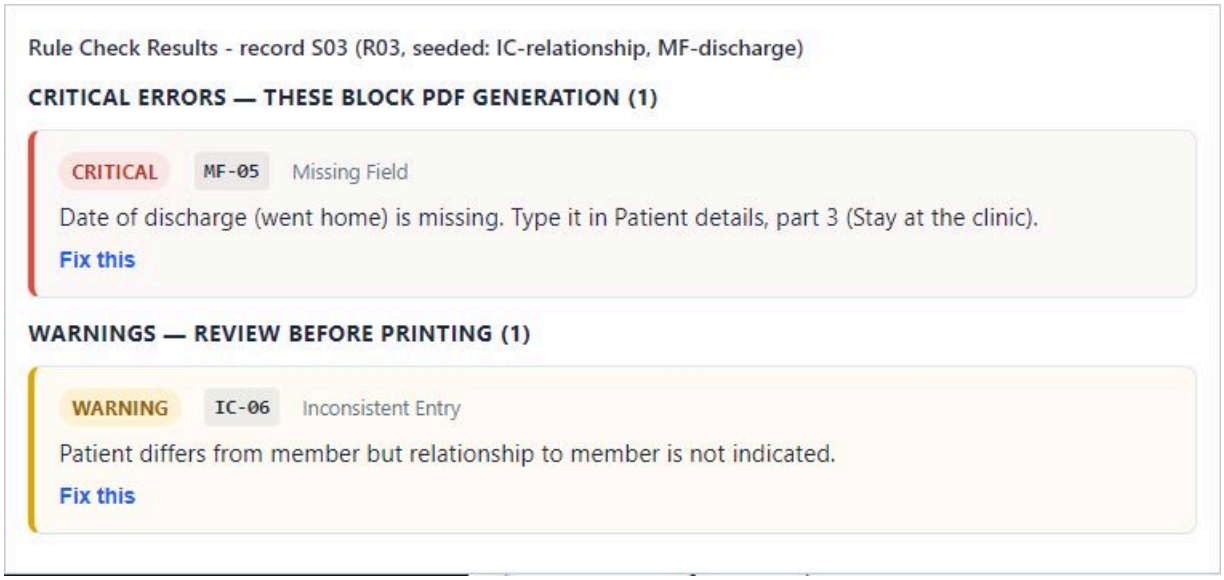
Fired: **MF-03, MF-06, SP-02**. Caught: **MF-03, SP-02**; also fired as a consequence: MF-06



Record S03 - patient R03 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
Unmarried patient filed under another member (her father), relationship to member left blank	memberIsPatient: ticked -> not ticked; memberFirstName: AILEEN -> ROLANDO; memberLastName: MANGAO -> MANGAO; memberSex: Female -> Male; civilStatus: Married -> Single; relationship: Self -> (blank)	IC-06	-
Date discharged left blank	dateDischarge: 2026-09-29 -> (blank)	MF-05	MF-06

Fired: **IC-06, MF-05**. Caught: **IC-06, MF-05**



Record S04 - patient R04 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
Logbook page failed the image quality check	image quality check: passed -> failed	IQ-01	-
Member PIN left blank	patientPIN: 99-123488376-4 -> (blank); memberPIN: 99-123488376-4 -> (blank)	MF-02	MF-06

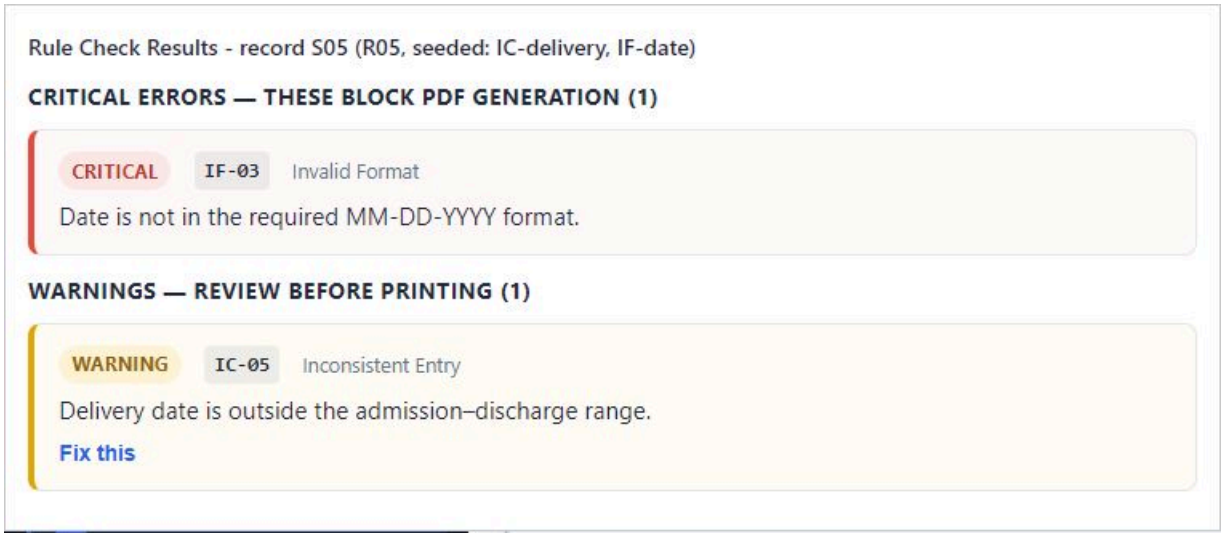
Fired: **IQ-01, MF-02**. Caught: **IQ-01, MF-02**



Record S05 - patient R05 (CSF, CF1, CF2, PMRF)

Seeded error	What was changed	Must fire	May also fire
Delivery date before admission	deliveryDate: 2026-08-29 -> 2026-08-28	IC-05	-
Birth date kept as unreadable text (OCR's letter O for a zero)	patientDOB: 2026-08-29 -> O8-29-2026	IF-03	-

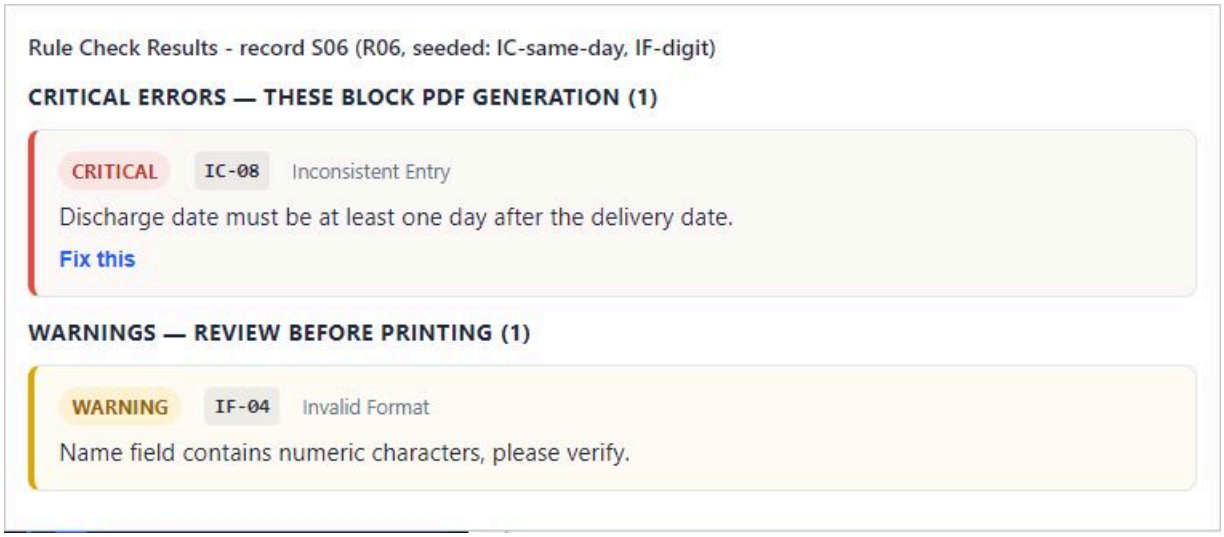
Fired: **IC-05, IF-03**. Caught: **IC-05, IF-03**



Record S06 - patient R06 (CSF, CF1, CF2, PMRF)

Seeded error	What was changed	Must fire	May also fire
Discharged the same day as the delivery	dateDischarge: 2026-09-07 -> 2026-09-06	IC-08	-
Digit in the patient's first name	patientFirstName: BABY BOY -> B1BY BOY	IF-04	-

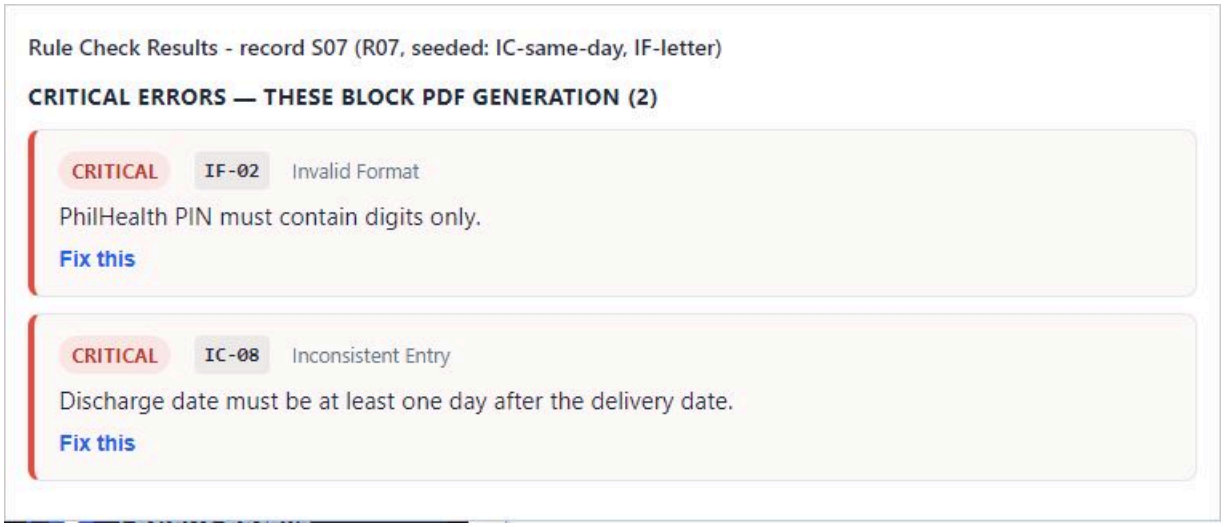
Fired: **IC-08, IF-04**. Caught: **IC-08, IF-04**



Record S07 - patient R07 (CSF, CF1, CF2, PMRF)

Seeded error	What was changed	Must fire	May also fire
Discharged the same day as the delivery	dateDischarge: 2026-09-29 -> 2026-09-28	IC-08	-
Letter in the PIN	memberPIN: 99-123512133-7 -> 99-12O3512133-7	IF-02	-

Fired: **IC-08, IF-02**. Caught: **IC-08, IF-02**



Record S08 - patient R08 (CSF, CF1, CF2)

Seeded error	What was changed	Must fire	May also fire
Patient name left blank	patientLastName: CAMINADE -> (blank); patientFirstName: MARY JOY -> (blank); patientMiddleName: SABANAL -> (blank); patientNameExt: (blank) -> (blank)	MF-01	MF-06
Date admitted left blank	dateAdmitted: 2026-08-29 -> (blank)	MF-04	MF-06

Fired: **MF-01, MF-04**. Caught: **MF-01, MF-04**



Record S09 - patient R09 (CSF, CF1, CF2)

Seeded error	What was changed	Must fire	May also fire
Member date of birth left blank	patientDOB: 1990-05-30 -> (blank)	MF-03	MF-06
Four required fields left blank (accommodation, disposition, HCI accreditation no., HCI name)	accommodation: Non-Private -> (blank); disposition: Improved -> (blank); hciPAN: H99999999 -> (blank); hciName: SAMPLE LYING-IN CLINIC -> (blank)	MF-06	-

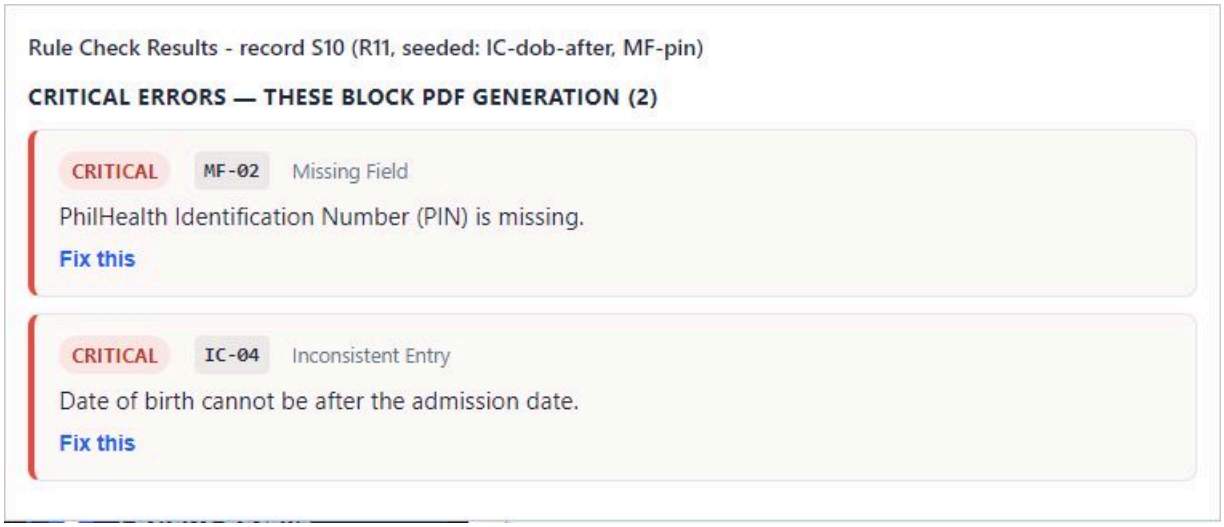
Fired: **MF-03, MF-06**. Caught: **MF-03, MF-06**



Record S10 - patient R11 (CSF, CF1, CF2, PMRF)

Seeded error	What was changed	Must fire	May also fire
Member born after admission	memberDOB: 1999-04-11 -> 2026-09-27	IC-04	-
Member PIN left blank	memberPIN: 99-123543809-1 -> (blank)	MF-02	MF-06

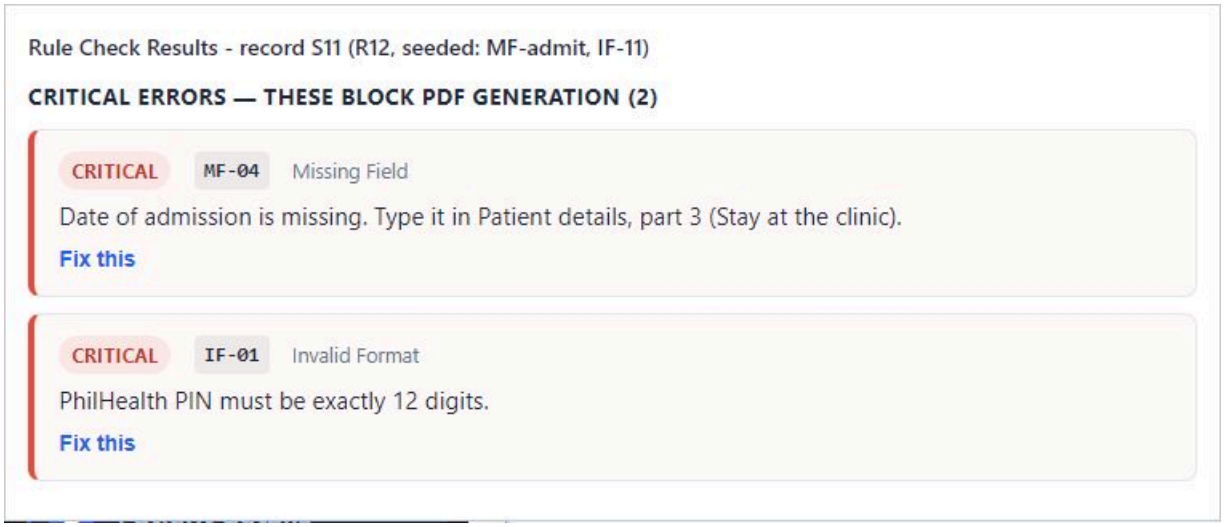
Fired: **IC-04, MF-02**. Caught: **IC-04, MF-02**



Record S11 - patient R12 (CSF, CF1, CF2, PMRF)

Seeded error	What was changed	Must fire	May also fire
Date admitted left blank	dateAdmitted: 2026-08-29 -> (blank)	MF-04	MF-06
PIN with 11 digits	memberPIN: 99-123551728-2 -> 99-12355172-8	IF-01	-

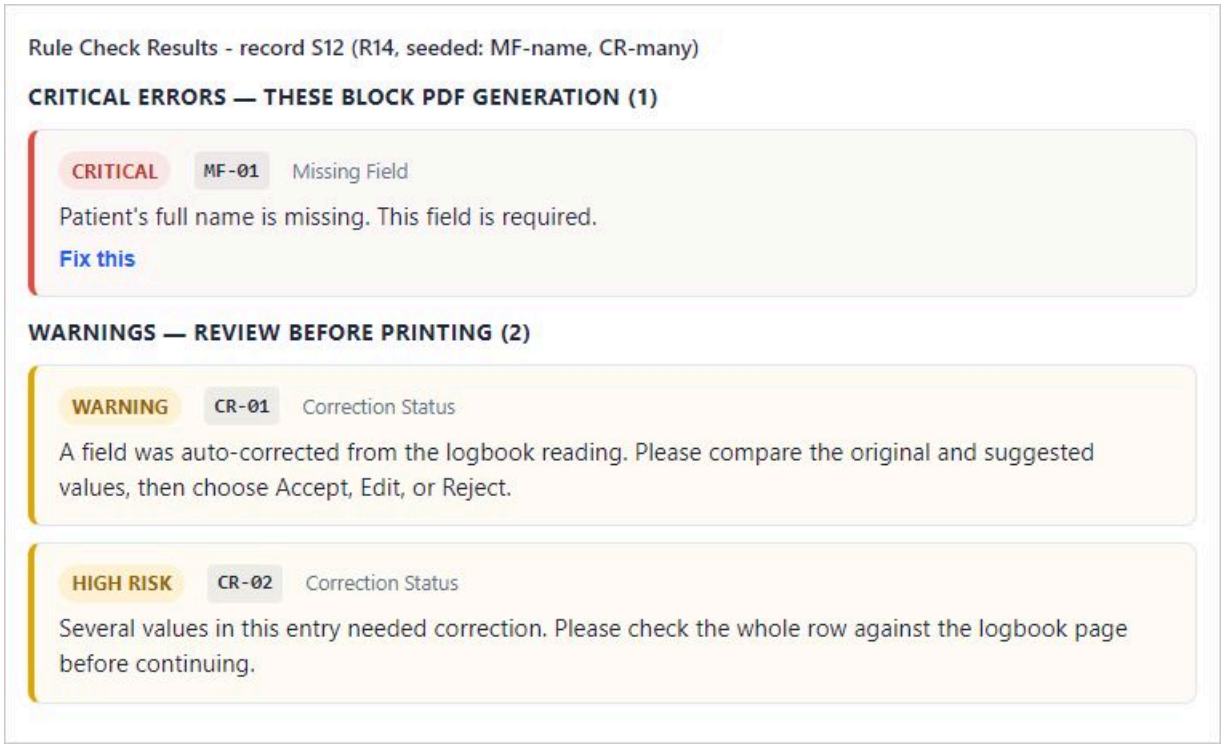
Fired: **IF-01, MF-04**. Caught: **IF-01, MF-04**



Record S12 - patient R14 (CSF, CF1, CF2)

Seeded error	What was changed	Must fire	May also fire
Patient name left blank	patientLastName: DAÑO -> (blank); patientFirstName: SHIELA MAE -> (blank); patientMiddleName: ALCUIZAR -> (blank); patientNameExt: (blank) -> (blank)	MF-01	MF-06
Five correction suggestions not yet reviewed	correction suggestions not reviewed: 0 -> 5	CR-02	CR-01

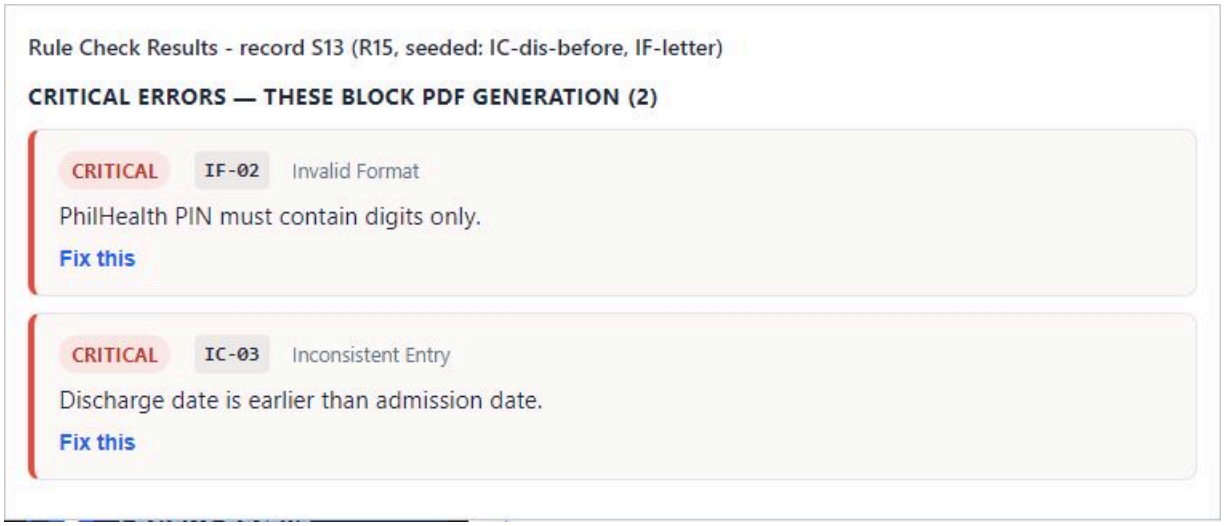
Fired: **CR-01, CR-02, MF-01**. Caught: **CR-02, MF-01**; also fired as a consequence: CR-01



Record S13 - patient R15 (CSF, CF1, CF2)

Seeded error	What was changed	Must fire	May also fire
Discharge date before admission	dateDischarge: 2026-09-04 -> 2026-09-02	IC-03	IC-05, IC-08
Letter in the PIN	patientPIN: 99-123575485-5 -> 99-12O3575485-5; memberPIN: 99-123575485-5 -> 99-12O3575485-5	IF-02	-

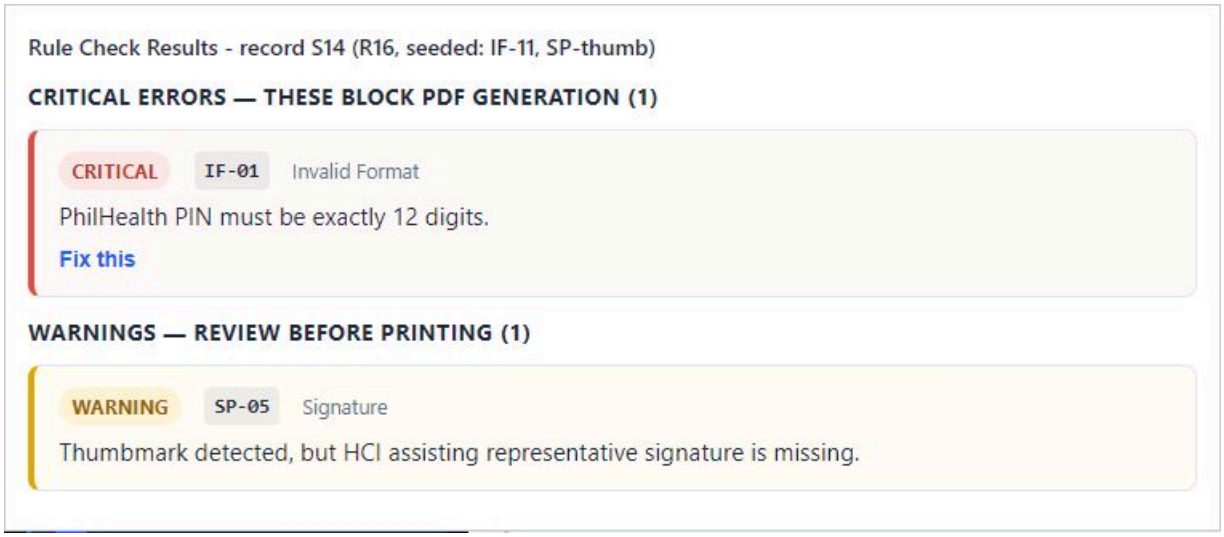
Fired: **IC-03, IF-02**. Caught: **IC-03, IF-02**



Record S14 - patient R16 (CSF, CF1, CF2)

Seeded error	What was changed	Must fire	May also fire
PIN with 11 digits	patientPIN: 99-123583404-6 -> 99-12358340-4; memberPIN: 99-123583404-6 -> 99-12358340-4	IF-01	-
Thumbmark without the HCI assisting representative	thumbmark signature: no thumbmark -> thumbmark used; hciAssist signature: present -> missing	SP-05	-

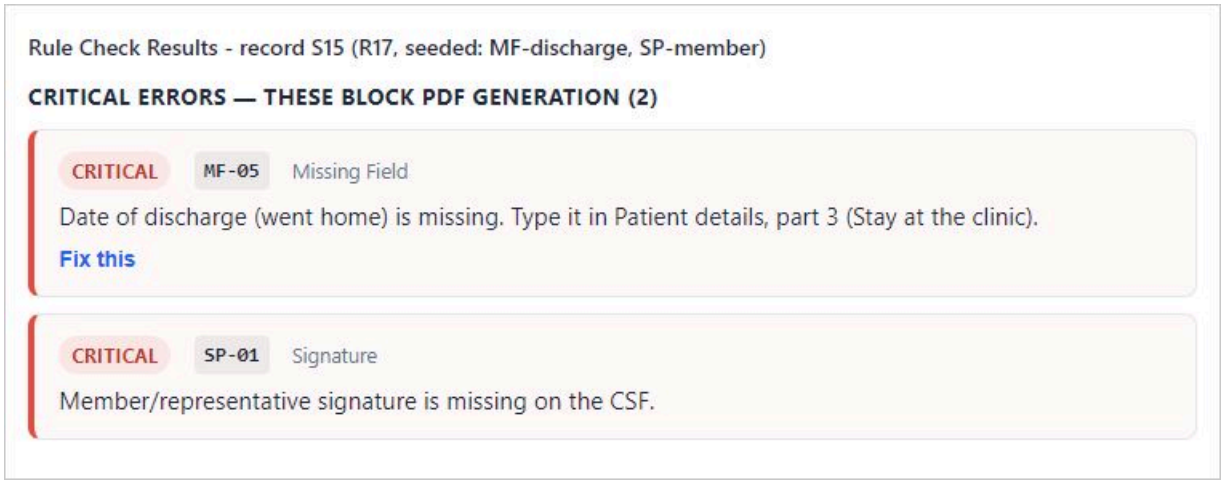
Fired: **IF-01, SP-05**. Caught: **IF-01, SP-05**



Record S15 - patient R17 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
Date discharged left blank	dateDischarge: 2026-08-27 -> (blank)	MF-05	MF-06
Member signature missing (signed form checked)	member signature: present -> missing	SP-01	-

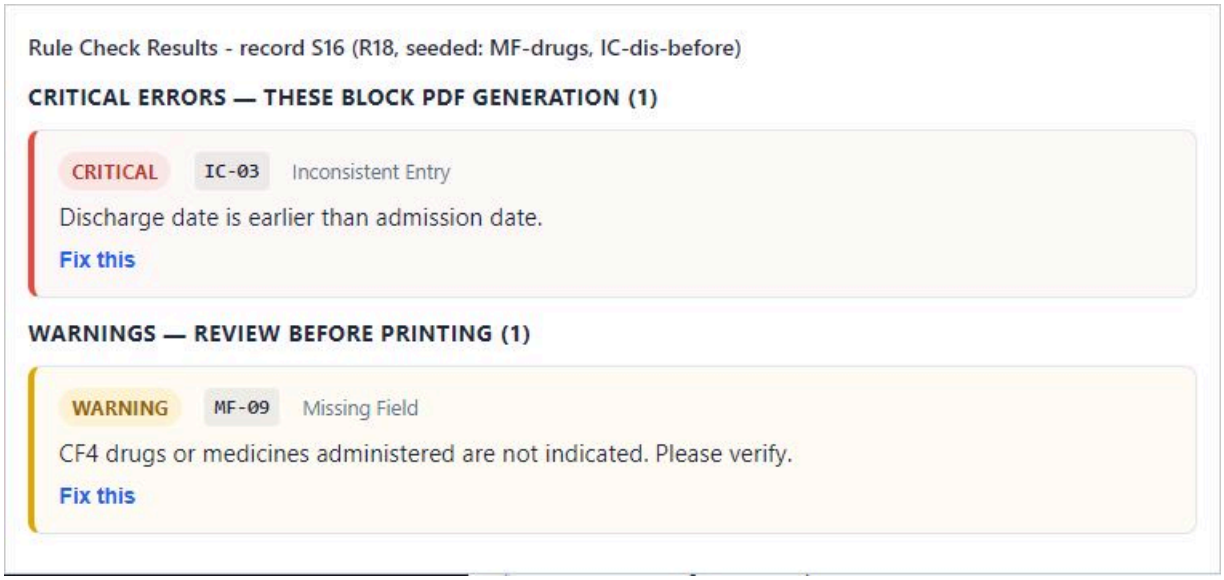
Fired: **MF-05, SP-01**. Caught: **MF-05, SP-01**



Record S16 - patient R18 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
CF4 drugs administered left blank	cf4DrugName1: COMBINED ORAL CONTRACEPTIVE PILLS -> (blank); cf4DrugName2: (blank) -> (blank); cf4DrugName3: (blank) -> (blank)	MF-09	MF-06
Discharge date before admission	dateDischarge: 2026-08-30 -> 2026-08-28	IC-03	IC-05, IC-08

Fired: **IC-03, MF-09**. Caught: **IC-03, MF-09**



Record S17 - patient R19 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
CF4 drugs administered left blank	cf4DrugName1: MEDROXYPROGESTERONE ACETATE (DMPA) -> (blank); cf4DrugName2: (blank) -> (blank); cf4DrugName3: (blank) -> (blank)	MF-09	MF-06
HCI representative signature missing	hciRep signature: present -> missing	SP-03	-

Fired: **MF-09, SP-03**. Caught: **MF-09, SP-03**



Record S18 - patient R20 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
Attending provider signature missing on CF4	cf4Provider signature: present -> missing	SP-06	-
Employer signature missing for an employed member	memberType: (blank) -> Employed Private; employer signature: present -> missing	SP-04	-

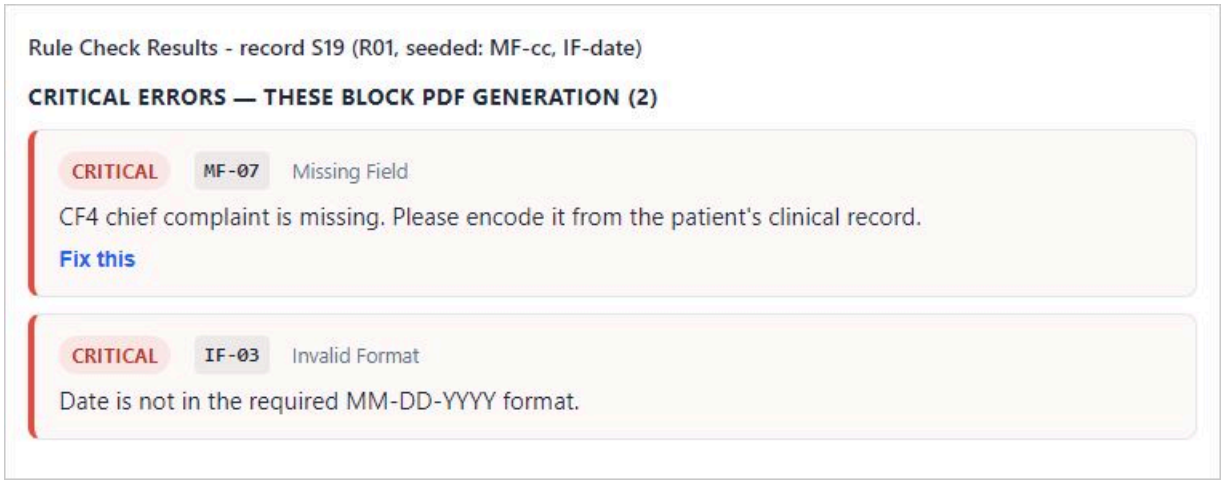
Fired: **SP-04, SP-06**. Caught: **SP-04, SP-06**



Record S19 - patient R01 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
CF4 chief complaint left blank	cf4ChiefComplaint: HYPOGASTRIC PAIN RADIATING AT THE ____ -> (blank)	MF-07	MF-06
Birth date kept as unreadable text (OCR's letter O for a zero)	patientDOB: 1993-01-09 -> O1-09-1993	IF-03	-

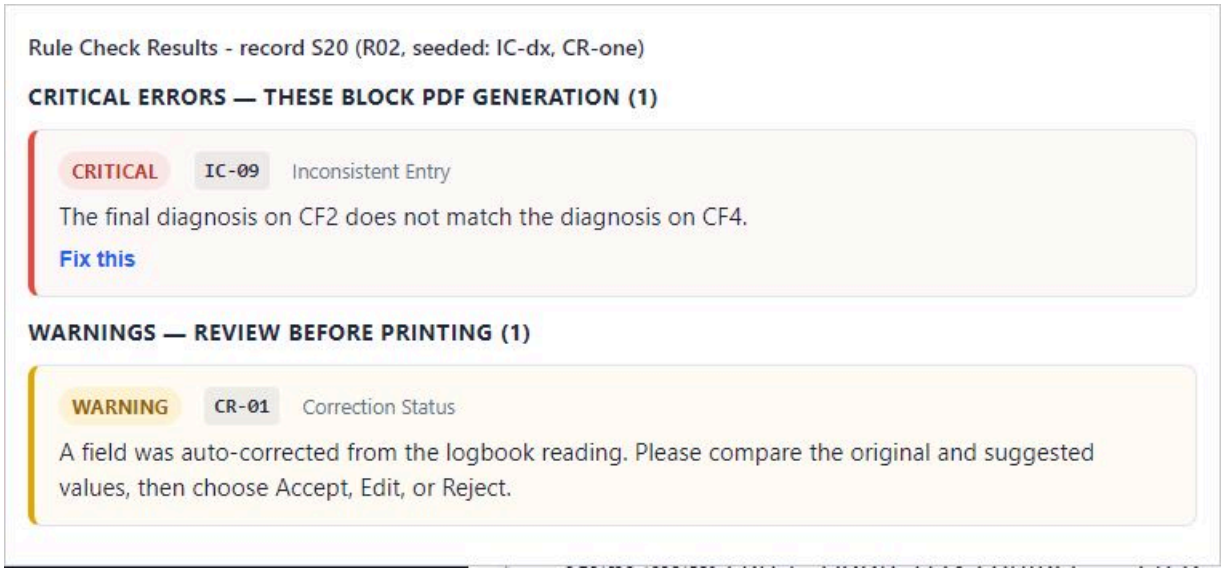
Fired: **IF-03, MF-07**. Caught: **IF-03, MF-07**



Record S20 - patient R02 (CSF, CF1, CF2, CF4)

Seeded error	What was changed	Must fire	May also fire
CF2 and CF4 diagnoses differ	dxADiagnosis: SPONTANEOUS VERTEX DELIVERY -> PREGNANCY UTERINE; cf4FinalDiagnosis: PUFT 39 WKS AOG DELIVERED VIA NSVD TO A LIVE TERM BABY GIRL AGA W/ AS OF 8,9 G2P2 (2002) -> ACUTE GASTROENTERITIS	IC-09	-
One correction suggestion not yet reviewed	correction suggestions not reviewed: 0 -> 1	CR-01	-

Fired: **CR-01, IC-09**. Caught: **CR-01, IC-09**



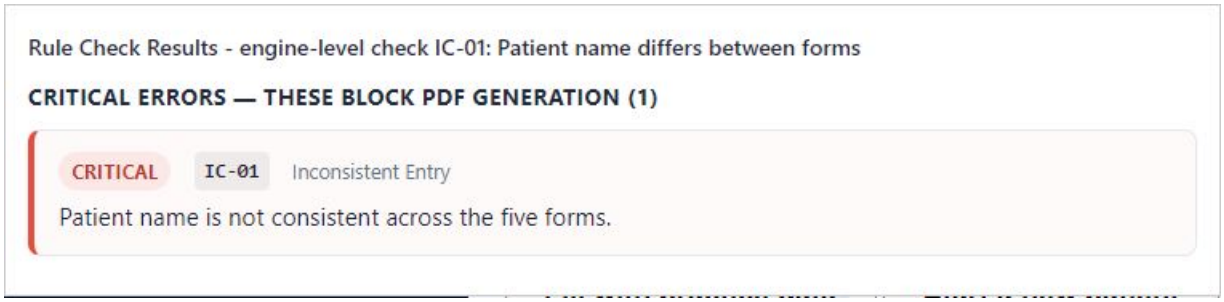
6. Engine-level checks (IC-01, IC-02, IC-07, MF-08)

These four rules cannot be reached on a real claim (section 3). Each is checked on a clean record (patient R01, with CF4): the engine's own computed features for the record, with only the feature the rule reads set the way the rule tests it; the rules then run and the page's own panel shows the result. The rule must fire, and nothing else. This checks the rule's logic; it is not part of the precision, recall, F1 or false alarm figures.

Rule	What it guards	Feature set	Rules fired on the record as it is	Rules fired with the feature set	New flag(s)	Fires, and only it
IC-01	Patient name differs between forms	{'name_match_across_forms': 0}	none	IC-01	IC-01	Yes
IC-02	PIN differs between CSF, CF4 and PMRF	{'pin_match_across_forms': 0}	none	IC-02	IC-02	Yes
IC-07	Admission/discharge dates differ between CSF, CF2 and CF4	{'dates_match_across_forms': 0}	none	IC-07	IC-07	Yes
MF-08	CF4 course in the ward blank	{'cf4CourseInWard_present': 0}	none	MF-08	MF-08	Yes

MF-08 on the real page. Every CF4 course-in-the-ward box was emptied; the page wrote the course back (cf4CourseDate2 = 2026-08-30; cf4CourseOrder2 = DISCHARGE AT 04:00 PM AND GIVEN DATES FOR POSTPARTUM FOLLOW UP SCHEDULES.), so the course in the ward can never be blank on a claim and MF-08 cannot fire there.

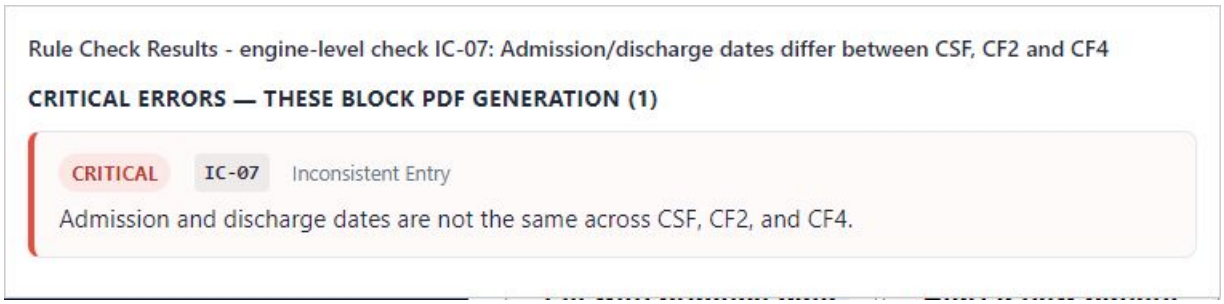
IC-01 - Patient name differs between forms: fires, and only it



IC-02 - PIN differs between CSF, CF4 and PMRF: fires, and only it



IC-07 - Admission/discharge dates differ between CSF, CF2 and CF4: fires, and only it



MF-08 - CF4 course in the ward blank: fires, and only it

Rule Check Results - engine-level check MF-08: CF4 course in the ward blank

CRITICAL ERRORS — THESE BLOCK PDF GENERATION (1)

CRITICAL `MF-08` Missing Field

CF4 course in the ward is missing. Please encode it from the patient's clinical record.

Fix this

7. The clean records

20 records with no seeded error: 0 flagged (none). Each panel below is the page's own result for that record.

Record C01 - patient R01: no flags

Rule Check Results - record C01 (R01, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C02 - patient R02: no flags

Rule Check Results - record C02 (R02, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C03 - patient R03: no flags

Rule Check Results - record C03 (R03, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C04 - patient R04: no flags

Rule Check Results - record C04 (R04, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C05 - patient R05: no flags

Rule Check Results - record C05 (R05, clean)

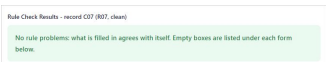
No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C06 - patient R06: no flags

Rule Check Results - record C06 (R06, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C07 - patient R07: no flags



Record C08 - patient R08: no flags

Rule Check Results - record C08 (R08, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C09 - patient R09: no flags

Rule Check Results - record C09 (R09, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C10 - patient R11: no flags

Rule Check Results - record C10 (R11, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C11 - patient R12: no flags

Rule Check Results - record C11 (R12, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C12 - patient R14: no flags

Rule Check Results - record C12 (R14, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C13 - patient R15: no flags

Rule Check Results - record C13 (R15, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C14 - patient R16: no flags

Rule Check Results - record C14 (R16, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C15 - patient R17: no flags

Rule Check Results - record C15 (R17, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C16 - patient R18: no flags

Rule Check Results - record C16 (R18, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C17 - patient R19: no flags

Rule Check Results - record C17 (R19, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C18 - patient R20: no flags

Rule Check Results - record C18 (R20, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C19 - patient R01: no flags

Rule Check Results - record C19 (R01, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

Record C20 - patient R02: no flags

Rule Check Results - record C20 (R02, clean)

No rule problems: what is filled in agrees with itself. Empty boxes are listed under each form below.

8. What this means

The validation engine flags every seeded error and nothing else. On 20 records with 40 seeded errors it flagged 40 (recall 1.0) with no false flags (precision 1.0), and it raised nothing on the 20 clean records (false alarm rate 0.0). 27 of the 31 rules were reached by seeded errors on real records; the other 4 cannot be reached on a real claim - the forms are filled from one entry, and the course in the ward is written by the page - and fire correctly when their condition is set at engine level.

Because the errors were designed from the rule definitions, this shows the rules work as written; how many real-world errors they catch is for the user sessions to show.